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METHOD OF STATEMENT FOR AIR DUCT PRESSURE LEAK TEST

Properad BY: _____
Subcontractor (Name & Signature) (Date)

Checked BY: _____
Project In- Charge (Name & Signature) (Date)

Approval BY: _____
Consultant (Name & Signature) (Date)

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1.0 Purpose

The purpose of this method statement is to define the systematic procedure for conducting air leakage testing on sheet metal ductwork to verify its integrity and ensure compliance with the allowable leakage rates specified in the project documents and relevant industry standards.

2.0 Scope of Work

This procedure is applicable to all low, medium, and high-pressure supply, return, and exhaust ductwork as identified in the project drawings. This test applies to the ductwork sections alone. All ancillary equipment, including but not limited to Air Handling Units (AHUs), fans, fire dampers, volume control dampers, silencers, and terminal units, shall be isolated from the section under test.

3.0 References

- **DW/143:** A Practical Guide to Ductwork Leakage Testing
- **DW/144:** Specification for Sheet Metal Ductwork
- **BS EN 12237:** Ventilation for buildings - Ductwork - Strength and leakage of sheet metal ducts
- Project ACMV Approved Drawings
- Workplace Safety and Health (WSH) Act

4.0 Responsibilities

Role	Key Responsibilities
Project Manager	Overall accountability for successful project execution. Ensures allocation of necessary resources (manpower, calibrated equipment) and acts as the primary liaison with the client for scheduling and issue resolution.
Site / Coordinator Supervisor	On-site authority for the test. Manages the Permit-to-Work, supervises the technical team, verifies all pre-test checks are complete, records all test data, and formally requests client/consultant witness inspections.
	Independent quality authority. Verifies material compliance and the validity of calibration certificates. Witnesses the test, validates results against acceptance criteria, and is

	responsible for compiling and submitting the final, comprehensive
WSH Supervisor	Enforces compliance with the approved Risk Assessment (RA) and Safe Work Procedure (SWP). Conducts pre-work toolbox talks, ensures the test area is correctly secured.
Skilled Technicians	Execute the physical work under direct supervision, including system filling, venting, pressurization, and depressurization. Responsible for the correct installation of test equipment and diligent monitoring.

5.0 Tools and Equipment

- Calibrated Duct Leakage Tester (e.g., PAN341 with TA465-P) with valid calibration certificate.
- Calibrated Micromanometer (e.g., PVM610).
- Flexible ducting for connecting the tester.
- Fan blower.
- Temporary blanking plates, inflatable bags, and heavy-duty tape for sealing openings.
- Approved duct sealant.
- Personal Protective Equipment (PPE): Safety glasses, gloves, safety helmet, safety Shoe, safety Vest.
- Ladders / Scaffolding / Mobile Elevating Work Platform (MEWP) as required.
- Hand tools (screwdrivers, hammers, Battery Drill etc.).

 Duct leak test Machine	 13 ft (4 m) long Ø4 in. (100 mm) plastic flexible duct	 16 ft (5 m) silicone tube (blue)
 TA465-P Multi-function Instrument	 PVM610 Micromanometer	 Duct Sealant
 PPE	 Ladder	 Tool box

6.0 Procedure

6.1 Pre-Test Preparation

1. Identify the section of ductwork to be tested as per the approved drawings. The test section should ideally not exceed 300m³ in of surface area to ensure manageable and accurate results.
2. We will only be tested on the main duct where pressure is the highest.
3. Calculate the total surface area in m² of the ductwork section to be tested.
4. Ensure the ductwork is clean, and all joint sealants (TDF corners, C-slip joints, etc.) are fully cured. Visually inspect all joints and seams for any obvious defects.
5. Excluding **damper, fire damper, air handling units' fans, heater, silencers, terminal boxes**, etc.
6. Ensure that the **Duct TDF or C-slip joint sections are not covered with fiberglass insulation**.
7. Ensure the duct system is sealed for at least 24 hours prior to testing.
8. Flow device as nozzle or flow Grid to select it.
9. Ensure test areas are cleared of any occupants not involved in testing, or setup signs to inform of risk of high-pressure tests.
10. It will be tested on the straight duct accessories will be mounted on the end or front of the duct.
11. Ductwork leakage testing will be carried out progressively, zone by zone.
12. Securely seal all openings within the test section. This includes open ends, branches, grille/diffuser collars, and connections to equipment. Use temporary blanking plates or inflatable bags. Tape up access doors.
13. Connect the leakage tester's flexible hose to the duct section via a securely attached temporary spigot.
14. Connect pressure sensing tubes to the ductwork at a suitable location, typically 2 to 5 meters away from the test rig connection point to avoid turbulence.

6.2 Test Execution

1. Confirm the correct power source is available for the test rig.
2. Zero the pressure sensors on the testing unit (TA465-P) with all ports open to the atmosphere.

3. Connect the Model TA465-P to the PAN341 by connecting the (+) and (-) ports on the Model TA465-P to the **P1 (+)** and **P2 (+)** ports located on the inside edge of the box shelf facing the fan

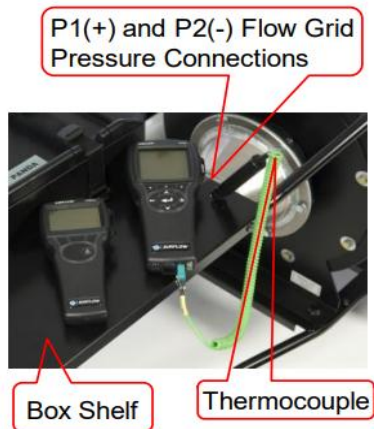


Figure 10. Connecting Instruments to PANDA Tester in High Flow Mode

4. On the test instrument, enter the required test data:
 - Duct Surface Area (m^2).
 - Ductwork Airtightness Class (e.g., Class A, B, or C, D as per DW/144 and project specification).
5. Turn on the tester fan. **Slowly and gradually** increase the fan speed to raise the pressure inside the duct.
6. Monitor the duct static pressure using the TSI PVM610. Do not exceed the target test pressure. The test pressure will be based on the ductwork classification:
 - **Class A:** 500 Pa
 - **Class B:** 1000 Pa
 - **Class C:** 2000 Pa
 - **Class C:** 2500 Pa
7. Once the target test pressure is reached, allow the pressure to stabilize for approximately 5 minutes.
8. After stabilization, record the final readings from the test unit. The key value is the **Leakage Rate (l/s)** or **Leakage Factor (l/s/m^2)** and m^3h . The unit will typically display a "Pass" or "Fail" result based on the entered data.
9. The test is always conducted using **positive pressure**, even for ductwork designed to operate under negative pressure.

6.3 Acceptance Criteria

- The test is considered successful if the measured leakage rate is **less than or equal to** the maximum permissible leakage rate calculated from DW/143 for the given pressure class and surface area.
- The result will be witnessed and signed off by the client/consultant's representative.
- After completing leakage testing for a section of duct, you can move onto the next section.

6.4 Procedure in Event of Failure

1. If the test fails, maintain pressure in the system (at a safe level) and conduct a thorough inspection of the duct section.
2. Listen for audible hissing and feel for air leaks at all joints, seams, and temporary seals. A non-toxic smoke generator can be used to help pinpoint leak locations.
3. Depressurize the system.
4. Repair all identified leaks using approved methods and sealants.
5. Allow the new sealant to fully cure.
6. Repeat the test procedure from step 6.2 until a "Pass" result is achieved.

7.0 Post-Test Activities

1. Once the test is accepted, safely depressurize the ductwork.
2. After the leak test is complete turn off the Leak test Machine Power.
3. Dismantle all testing equipment.
4. Carefully remove all temporary blanking plates, seals, and tape.
5. Patch the holes used for the test rig connection and pressure sensing tubes using approved methods.
6. Prepare and submit the formal Duct Leakage Test Report for approval. The report shall include project details, duct section identification, surface area, pressure class, test pressure, allowable leakage rate, measured leakage rate, and a Pass/Fail result, signed by both the testing lead and the witness.

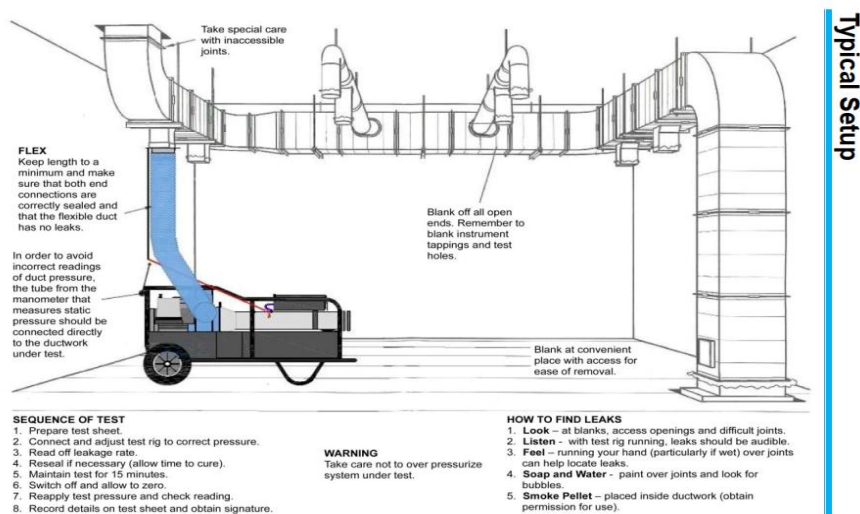
8.0 Documentation and Handover

Throughout the process, record all readings, observations, and test results on the Commissioning Test Sheet for the system.

1. Calibration certificates for all testing equipment.
2. Any failures, discrepancies, or non-conformities shall be recorded on a "Punch List" for rectification.
3. Upon successful completion of all tests, the Commissioning Engineer and Supervisor will sign the test sheet, confirming the System is safe and fully functional.
4. The completed and signed documentation will be included in the final Testing & Commissioning report for handover.
5. Compile the report for submission and verification with Maincon M&E / RTO or Consultant.
6. A formal handover of the commissioned system will be conducted with the client and facilities management team, including a demonstration of the system's operation.

9.0 Health and Safety

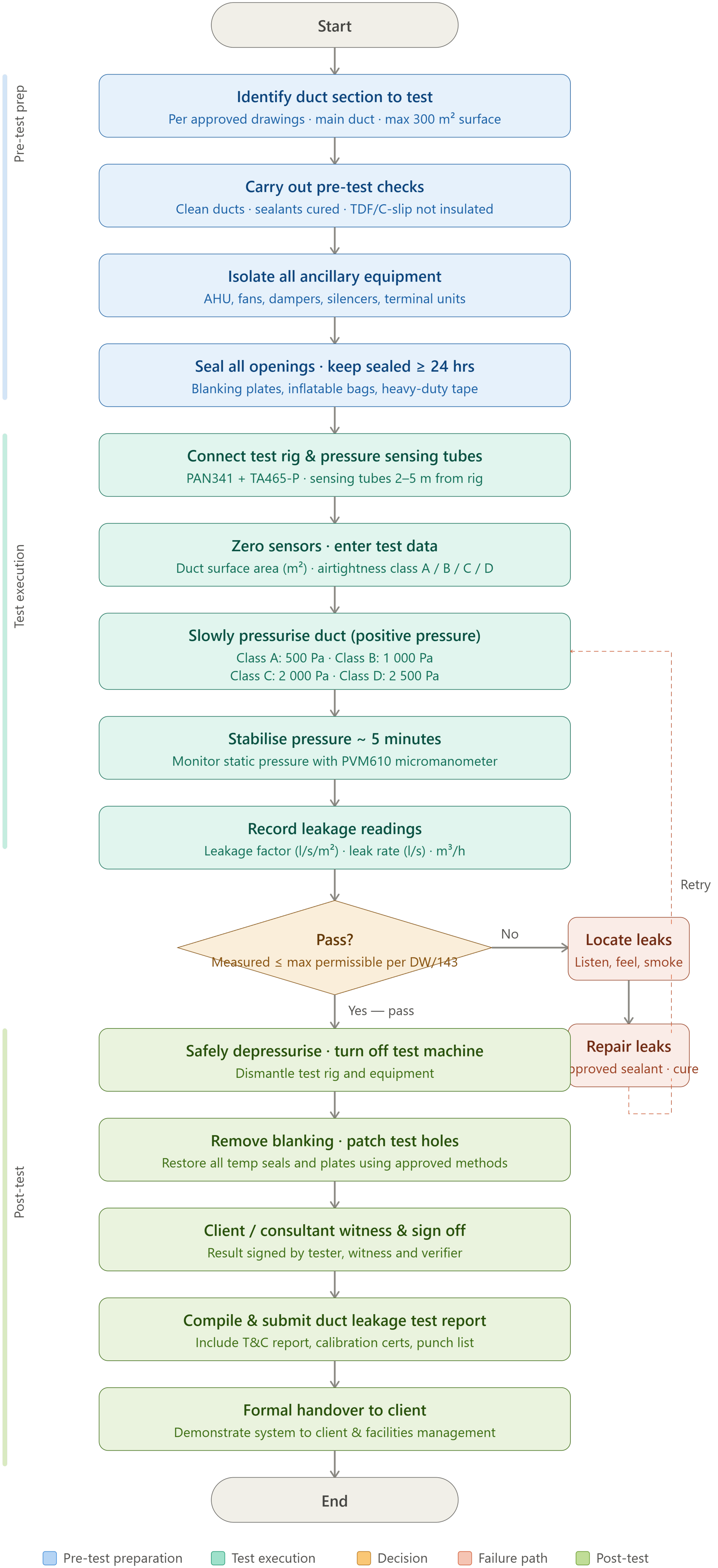
1. A risk assessment shall be conducted prior to the commencement of work.
2. All personnel involved must be trained and competent.
3. The main ACMV system connected to the duct section under test must be de-energized and placed under a **Lock-Out, Tag-Out (LOTO)** system.
4. The test area shall be cordoned off with warning signs indicating "Duct Pressure Test in Progress – Do Not Enter."
5. Ensure stable and secure access (ladders, scaffolds) for all works at height.
6. All temporary seals and blanking plates must be securely fastened to prevent them from becoming projectiles under pressure.



Reference Testing Photos



Duct Leak Inspection with RTO & Consultant.



Part One – A practical guide to ductwork leakage testing

1 GENERAL

With regard to air leakage, the responsibility for ensuring the achievement of a satisfactory job is divided between the Drawing Office, the Factory and the Site Erection Operatives. It is essential that there is full co-operation between them.

2 THE DRAWING OFFICE

The drawing office must:

2.1 Establish with the client or his representatives the class of ductwork called for in the job specification, i.e.:

- Class A: up to 500 Pa positive
- Class B: up to 1000 Pa positive
- Class C: up to 2000 Pa positive
- Class D: up to 2500 Pa positive

2.2 Establish with the client or his representative that the required leakage rate or rates are for the ductwork alone i.e. excluding dampers, fire dampers, air handling units, fans, heater batteries, silencers, terminal boxes, etc.

Additionally, establish the type of gasket material required in relation to the pressure class/velocity of the system.

2.5 Arrange for the supply of suitable medium, e.g. heavy-duty polythene

2.6 Make sure that all test points and devices can be reached with the minimum difficulty after the ductwork has been

2.7 To ensure that a reading can be taken plan test sections to have a permitted leakage approximately 25 per cent less than the volume generated by the test rig at the pressure required for each section.

2.8 Provide the erectors with details of test zones, duct operating pressures; indicate the nature of blanking devices, gasket material and tools to be used.

2.9 Prepare test sheets giving the information called for on the sample test sheet on page 9.

This information should indicate the content, the surface area of the ductwork tested, and the permitted loss for each section to be tested (this loss is to be expressed in per second per square metre of the ductwork).

SCHEDULE OF CHILLED WATER AIR HANDLING UNITS																															
S/N	Unit Designation	Unit Location		Make	Country of Manufacturing	Model	Type	Specified Capacity		Offered Capacity		On-Coil		Off-Coil		Chilled Water		Water Flow Rate	Water Pipe Size	Water Pressure Drop	Coil Face Velocity	Supply Air		Fresh Air	ESP	Calculated ESP	Dimension (WxLxH)				
		Storey	Area Served					Total	Sensible	Total	Sensible	DB	WB	DB	WB	In	Out					(l/s)	(Ø mm)					(kPa)	(m/s)	(cmh)	(cmh)
1	AHL-1.3-3-4	L3	Temp Ward	SAVER	Malaysia	AH(M)748H-2053W	Horizontal	315.0	110	195.6	101.19	27.77	22.67	14.60	14.49	8.50	13.50	9.34	100	35.85	2.51	22000	22000	10660	750	<737	2053x4481x1848				
2	AHL-1.3-3-4	L3	Temp Ward	SAVER	Malaysia	AH(M)748H-2053W	Horizontal	315.0	110	140.9	82.49	26.10	21.00	14.60	14.45	8.50	13.50	6.73	100	20.01	2.35	20570	20570	6180	750	737	2053x4481x1848				
3	AHL-1.3-3-5	L3	Temp Ward	SAVER	Malaysia	AH(M)748H-2053W	Horizontal	315.0	110	195.6	101.19	27.77	22.67	14.60	14.49	8.50	13.50	9.34	100	35.85	2.51	22000	22000	10660	750	<621	2053x4481x1848				
4	AHL-1.3-3-5	L3	Temp Ward	SAVER	Malaysia	AH(M)748H-2053W	Horizontal	315.0	110	93.04	54.6	26.10	21.00	14.60	14.47	8.50	13.50	4.45	100	11.56	1.51	12662	12662	4480	750	621	2053x4481x1848				

- d. Leakage class as A, B, C, or D. Note that tests with negative pressures must be selected as negative tests, as indicated by -.
 - e. Test Length, or duration of leakage test, usually 5 minutes.
2. Increase the blower speed until the desired static pressure is achieved.
 3. When the static pressure has stabilized, select **Run Test** and press $\leftarrow$.
 4. The display will show the readings on the right. Leakage Factor and Leak Rate will update in real time, while other parameters will remain constant.
If the Leakage Factor and Leak Rate are sufficiently stable, press the **START** soft key or the

<u>LEAKAGE TEST</u>	
Leakage Factor	x.xx
Leak Limit	x.xx
Leak Rate	x.xx
Status	OK
Flow Device	Flow Grid
Baro Pressure	20.20
Temperature	20°C
Time	9:55
Standard	TestXXX
	Sample
	0
(Sample Saved 1)	

EU Standards

Ductwork classification and maximum air leakage. Note that EN1507, EN12237 Eurovent 2/2 and DW/143 all have the same formula to determine f_{max} , the Air Leakage Limit, although DW/143 uses units of l/s/m² whereas others use m³/s/m².

- EN 1507 (rectangular ductwork)

Air Tightness Class	Air Leakage Limit (f_{max}) m ³ /s/m ²	Static Pressure Limit (p_s) Pa			
		Negative	Positive at pressure class		
			1	2	3
A	$\frac{0.027 * p_t^{0.65}}{1000}$	200	400		
B	$\frac{0.009 * p_t^{0.65}}{1000}$	500	400	1000	2000
C	$\frac{0.003 * p_t^{0.65}}{1000}$	750	400	1000	2000
D*	$\frac{0.001 * p_t^{0.65}}{1000}$	750	400	1000	2000

* Class D ductwork is only for special apparatus

9.3. Testing Equipment Calibration Certificate



GRACECAL PTE LTD (GST Reg No: 201611630E)

31 Woodlands Close #02-05 Woodlands Horizon Singapore 737855
Tel: (65) 6742 3850 | Fax: (65) 6742 3849 | Email: sales@gracecal.com
Website: www.gracecal.com

CALIBRATION CERTIFICATE

CLIENT : **WINNER ENGINEERING PTE LTD**
18 Woodlands Industrial Park E1
Singapore 757738

CERTIFICATE NO : GCC241442
ISSUE DATE : 06-SEP-2024

INSTRUMENT : **MULTI FUNCTION ANEMOMETER**
MAKE : TSI
MODEL NO. : TA465-P-NB
PART NO. : ---
SERIAL NO. : TA4652029012

WORKS ORDER NO : GWO241601
DATE RECEIVED : 02-SEP-2024

DATE CALIBRATED : 06-SEP-2024
DUE DATE : 06-SEP-2025

AMBIENT TEMPERATURE:
(23 ± 5) °C
RELATIVE HUMIDITY:
(55 ± 10) % r.h.

RANGE : -----
GRADUATION : -----
(TAG NO.) : -----
PAGE : 1 of 2
STATUS : AS FOUND

The described instrument has been calibrated at Laboratory under the ambient conditions stated above.

This certificate provides traceability of measurement to the International System of Units (SI) and/or to units of measurement realized at the National Metrology Centre (NMC), Singapore or other recognized national metrology institutes.

The Instrument was calibrated by comparison with laboratory calibration testing equipment.

1. Gas Deadweight Pressure Tester S/N: 558735

Calibration Method

Calibration Method is in accordance with ISO/IEC 17025. The calibration method was carried out according to Technical Calibration Procedure AS-TM-M03 as a guide.

Calibration Results

1. The results of calibration are given on the attached calibration data sheet(s).
2. The expanded uncertainty of measurement associated with the calibration is estimated at a confidence level of approximately 95% with a coverage factor of k=2.00.
3. The user should determine the suitability of the instrument for its intended use.

Calibrated By:

PREM

ID: 748656

Approved By:

KUMAR

ID: 748670





GRACECAL PTE LTD (GST Reg No: 201611630E)

31 Woodlands Close #02-05 Woodlands Horizon Singapore 737855
Tel: (65) 6742 3850 | Fax: (65) 6742 3849 | Email: sales@gracecal.com
Website: www.gracecal.com

CALIBRATION CERTIFICATE

CERTIFICATE NO : GCC241442
ISSUED DATE : 06-SEP-2024

WORKS ORDER NO : GWO241601
PAGE : 2 of 2

APPLIED ACTUAL VALUE (Pa)		MEAN INSTRUMENT READING (Pa)				
		Ascending			Descending	
	Before Adj.	After Adj.	Error	Before Adj.	After Adj.	Error
Pressure Measurement						
0.00	0.1	---	-0.10	0.0	---	0.00
50.46	50.7	---	-0.24	50.7	---	-0.24
477.53	477.8	---	-0.27	477.8	---	-0.27
916.92	917.2	---	-0.28	917.2	---	-0.28



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Website: www.gracecal.com

CALIBRATION CERTIFICATE

CLIENT : **WINNER ENGINEERING PTE LTD**
18 Woodlands Industrial Park E1
Singapore 757738

CERTIFICATE NO : GCC241440
ISSUE DATE : 06-SEP-2024

INSTRUMENT : **MICROMANOMETER**
MAKE : AIRFLOW
MODEL NO. : PVM610
PART NO. : ---
SERIAL NO. : PVM612031009

WORKS ORDER NO : GWO241601
DATE RECEIVED : 02-SEP-2024

DATE CALIBRATED : 06-SEP-2024
DUE DATE : 06-SEP-2025

AMBIENT TEMPERATURE :
(23 ± 5) °C
RELATIVE HUMIDITY:
(55 ± 10) % r.h.

RANGE : ---
(TAG NO.) : ---

PAGE : 1 of 2
STATUS : AS FOUND

The described instrument has been calibrated at Laboratory under the ambient conditions stated above.

The reference measurement standards used are traceable to National Metrology Centre (**NMC**), Singapore or National Physical Laboratory (**NPL**), UK or National Institute of Standards and Technology (**NIST**), USA or other recognized National or International Standards Laboratories.

Calibration Method

Calibration Method is in accordance with **ISO/IEC 17025**. Calibration Method was carried out according to in-house Technical Calibration Procedure AS-TM-MS07, M03 specification as a guide.

The Instrument was calibrated by comparison with laboratory calibration testing equipment.

Instrument (Serial No): Gas Deadweight Pressure Tester S/N: 558735

Calibration Results

1. The results of calibration are given on the attached calibration data sheet(s).
2. The expanded uncertainty of measurement associated with the calibration is estimated at a confidence level of approximately 95% with a coverage factor of $k=2.00$.
3. The user should determine the suitability of the instrument for its intended use.

Calibrated By:

PREM

ID: 748656

Approved By:

KUMAR

ID: 748670





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Tel: (65) 6742 3850 | Fax: (65) 6742 3849 | Email: sales@gracecal.com
Website: www.gracecal.com

CALIBRATION CERTIFICATE

CERTIFICATE NO : GCC241440
ISSUED DATE : 06-SEP-2024

WORKS ORDER NO : GW0241601
PAGE : 2 of 2

	APPLIED ACTUAL VALUE (Pa)			MEAN INSTRUMENT READING (Pa)		
	Before Adj.	Ascending After Adj.	Error	Before Adj.	Descending After Adj.	Error
Pressure Measurement						
0.00	0.0	---	0.00	0.0	---	0.00
50.33	49.8	---	0.53	49.8	---	0.53
498.61	498.1	---	0.51	498.1	---	0.51
909.11	908.6	---	0.51	908.7	---	0.41